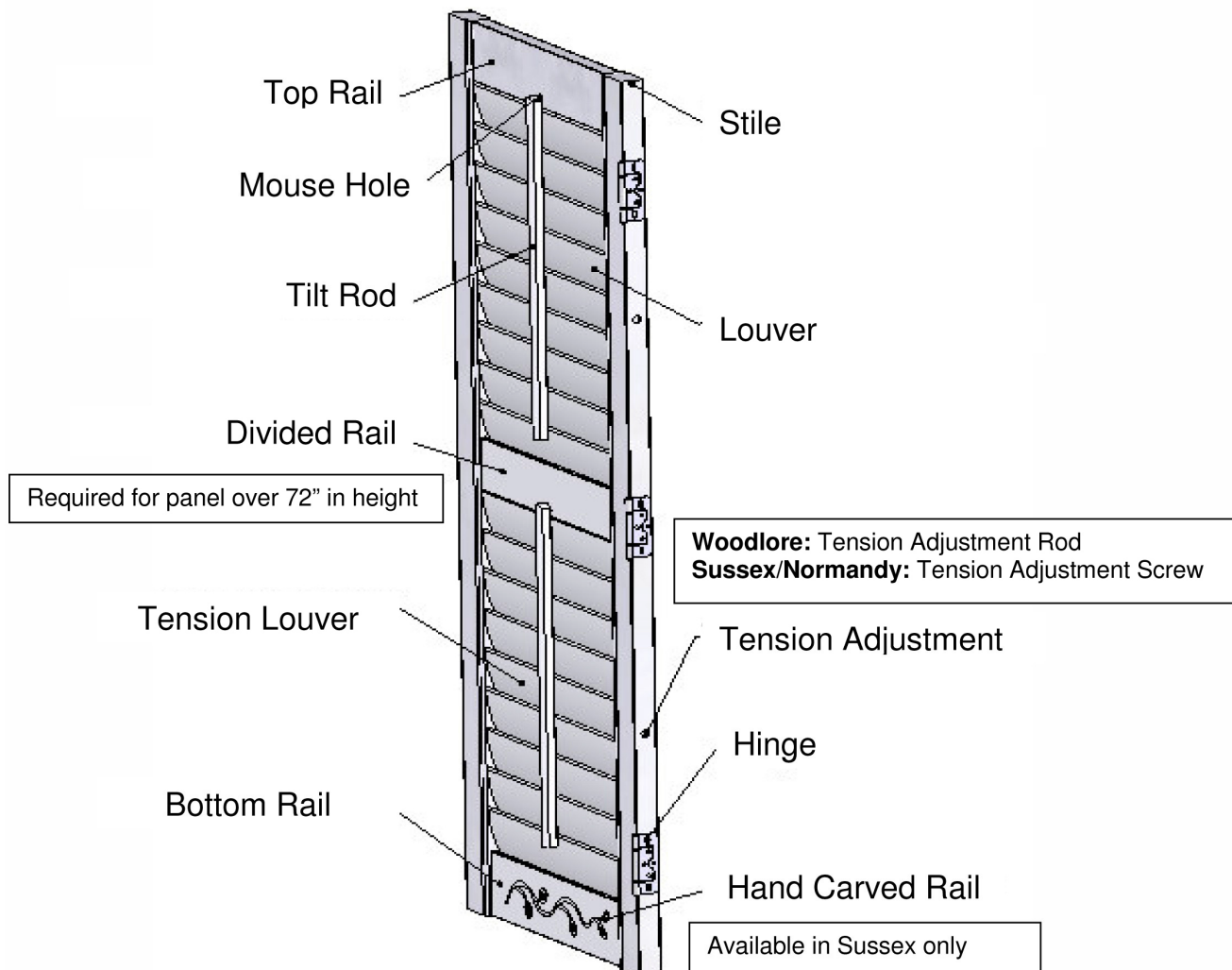


Measuring Guidelines

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• Standard Deduction	
• Café Shutter	
• Corner Window	
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• T Post	
• Other Window Situation	

Panel Anatomy



Basic Measuring

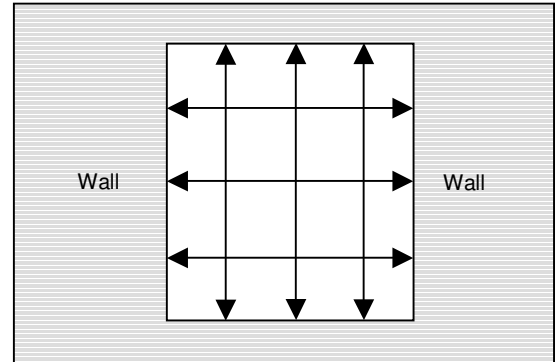
To get a precise and accurate measurement is the first step to a perfect shutter.

- ◇ Always use a metal measuring tape.
- ◇ Always measure to the closest $\frac{1}{16}$ ".

To measure your window

Width: Take 3 measurements at the top, middle and bottom of the window opening.

Height: Take 3 measurements at the left, center and right of the window opening.



Mounting Options

Inside Mount (IM)

- Shutters will fit inside a window opening.
 - There must be enough unobstructed depth on the mounting surface inside the window opening.
- For **IM**, please provide the **SMALLEST** height and width measurement among the three.

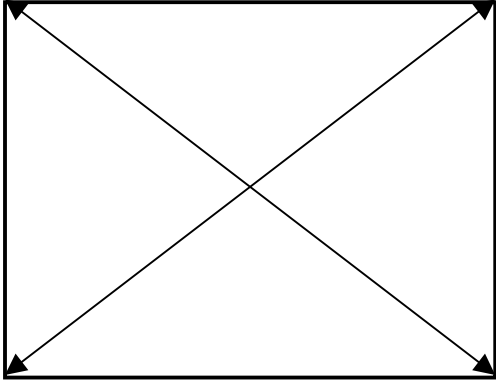
Outside Mount (OM)

- Shutters will fit outside a window opening.
 - There must be sufficient flat mounting surface around the window opening.
- For **OM**, please provide the **LARGEST** height and width measurement among the three.

Tip: When there is enough flat mounting surface, Outside Mount is generally a preferred method as it is more forgiving and less challenging compare to Inside Mount.

Basic Measuring

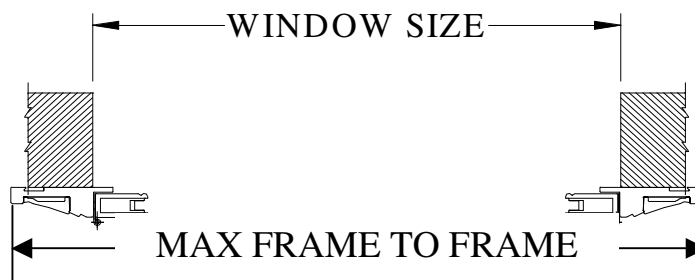
Determine Window Squareness



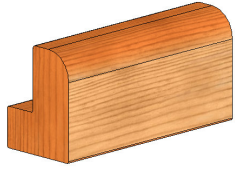
- Measure diagonally with a metal measuring tape
- Should the difference between the two diagonal measurements be greater than $\frac{1}{4}$ ", OM is highly recommended.

Measurement Type

- **Max Frame to Frame (MFF)**
Finished size of the shutters, measure from the outside edge of the left to right and the top to bottom frame.
- **Window Size (WS)**
Exact size of the window opening



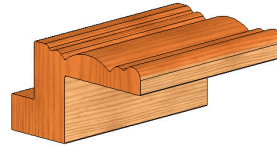
General Frame Categories



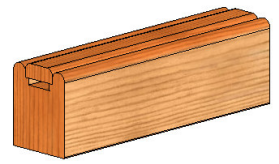
L Frame



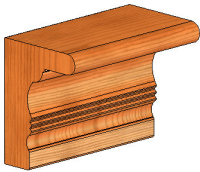
Deco Frame



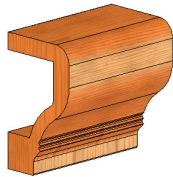
Z Frame



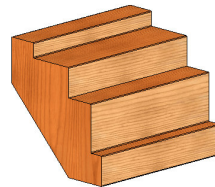
Hang Strip



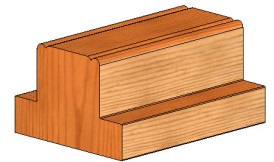
Sill Frame



Designer Sill Cap

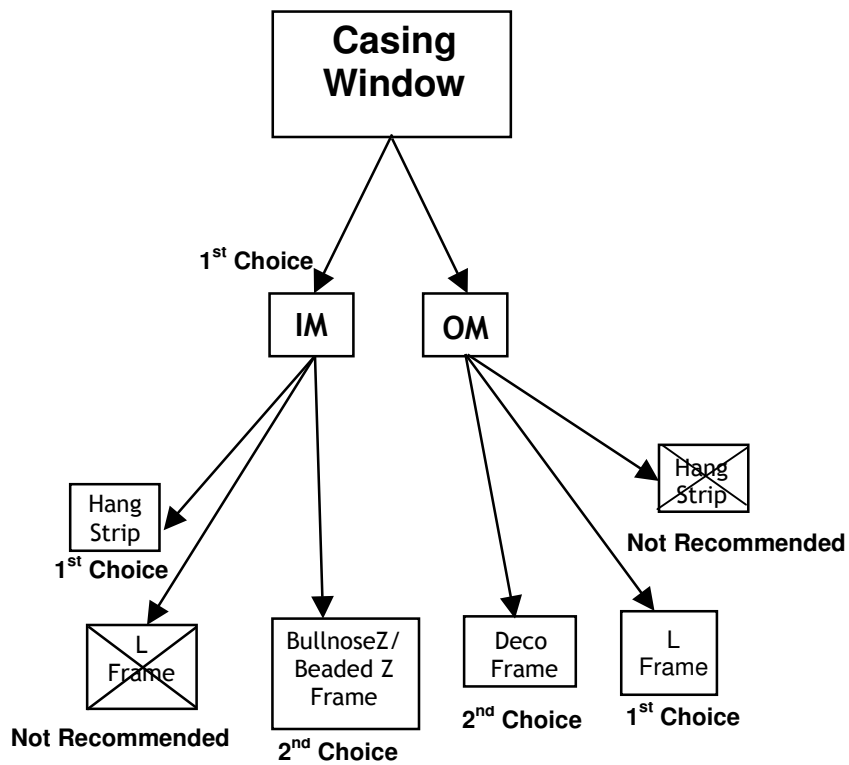
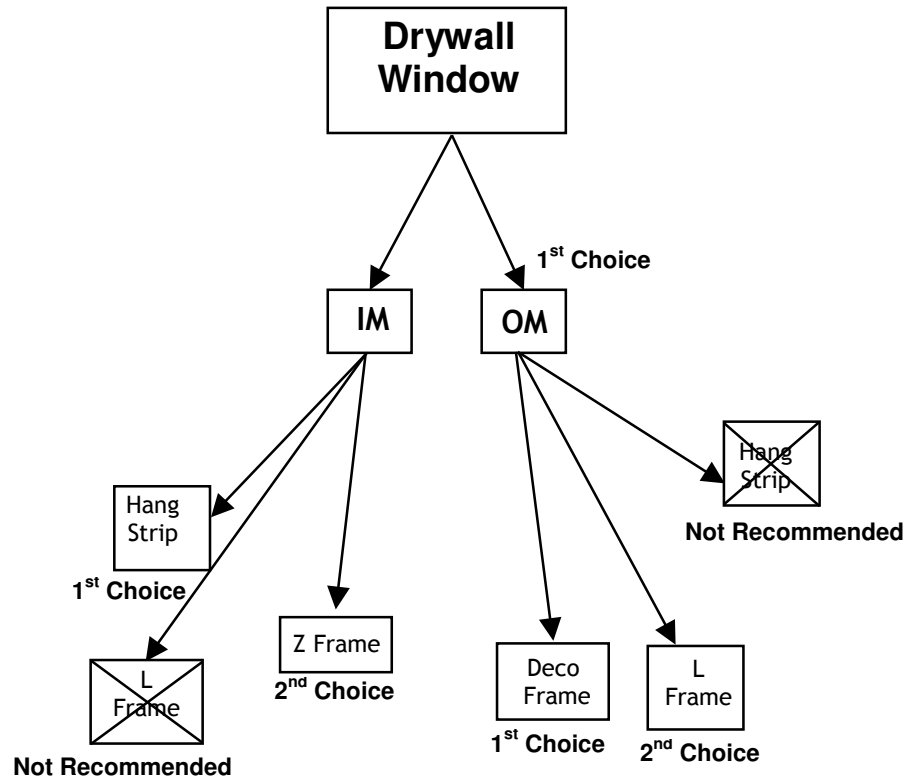


Corner Post

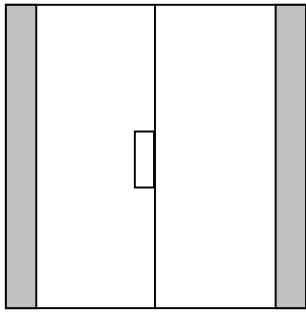


T Post

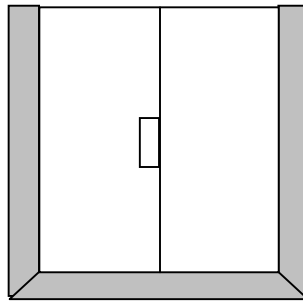
Frame Decision Tree



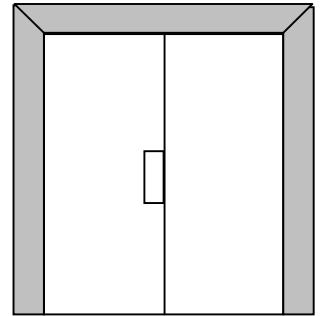
Number of Sides



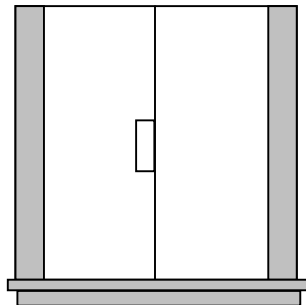
2-sided



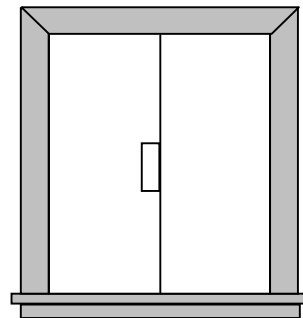
3-sided



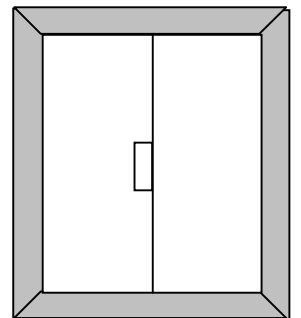
3-sided



2-sided + Deco Sill Frame/
Designer Sill Cap



3-sided + Deco Sill Frame/
Designer Sill Cap



4-sided

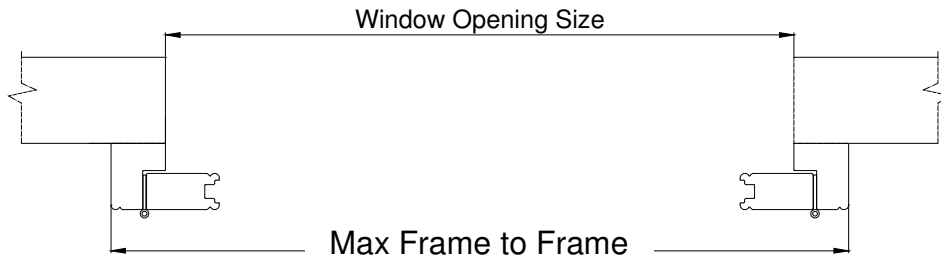
Measuring For Different Frames

L Frame

- OM or IM

OM

- The edge of L frame usually aligns with the window edge.
Under situation where there is moldings around the window, L frame can be mounted on the outside edge of the molding or on top of the molding.



IM

- Max Frame to Frame measurement is **required** for L frame inside mount.
- $\frac{1}{16}$ " is the suggested deduction on each side.



Feature:

- *L Frame OM leaves more room on the inside of the window opening as it protrudes more into the room compares to other frames.*

Note:

For 2-sided L Frame (IM & OM)
other than café shutter,
standard deduction for

Frame Height = $\frac{1}{16}$ "
Panel Height = $\frac{1}{4}$ "

unless specify otherwise.

Measuring For Different Frames

Hang Strip

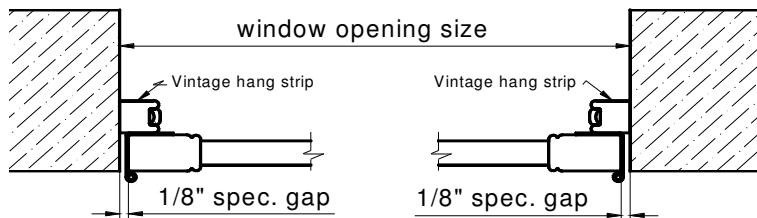
- OM or IM

IM

- Hang strip is placed behind the panel.
- Shutters come with two pre-drilled hang strips only. Two light blocks will be provided for the top and bottom of the window opening.

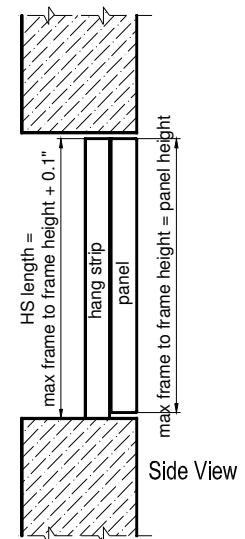
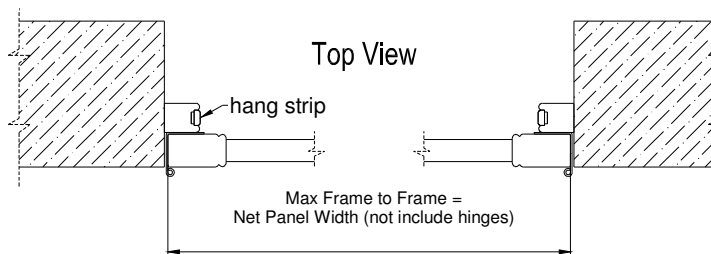
- **Window size measurement**

- ◇ Standard deduction for gap between panel and window casing = $\frac{1}{8}$ " for width, $\frac{1}{8}$ " to $\frac{3}{16}$ " for height.



- **Max Frame to Frame measurement**

- ◇ Max Frame to Frame width = panel width (not include hinges).
- ◇ Max Frame to Frame height = panel height
- ◇ The total length for hang strip will be $\frac{1}{10}$ " more than panel height.

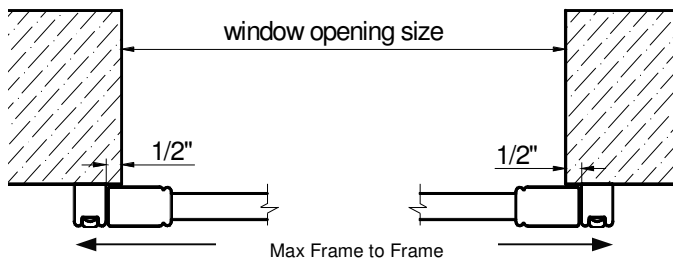


Measuring For Different Frames

Hang Strip

OM

- Hang strip is placed on the side of the panel
- Panel is $\frac{1}{2}$ " wider than the window opening on both left and right side



**Hang strip OM; beside the panels
(shown in Vintage Hang Strip)**

Note:
For 2-sided Hang Strip (OM)
other than café shutter,
standard deduction for

Frame Height = $\frac{1}{16}$ "
Panel Height = $\frac{1}{4}$ "

unless specify otherwise.

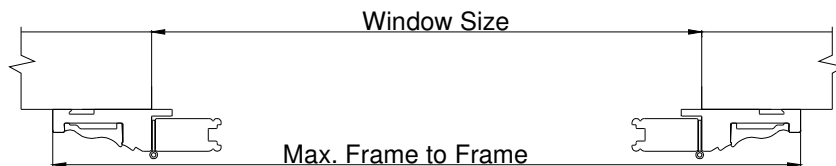
Feature:

- *An innovative frame design for easier installation. Hang strip comes with pre-drilled holes and can be installed with screws that will be hidden under the inserts.*

Measuring For Different Frames

Deco Frame

- OM only
- Must have enough flat surface around the window opening for frame installation.



Note:
For 2-sided Deco Frame (OM)
other than café shutter,
standard deduction for

Frame Height = $\frac{1}{16}$ "
Panel Height = $\frac{1}{4}$ "

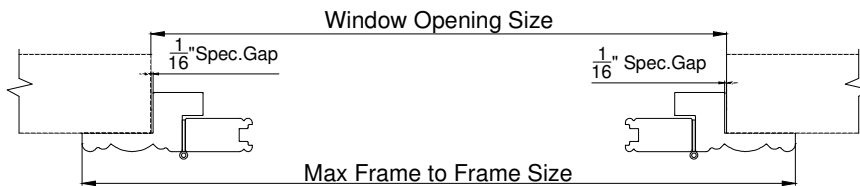
unless specify otherwise.

Feature:

- *Deco frame can help to hide an imperfect window.*
- *The elegant frame profile adds a decorative touch to an ordinary window.*
- *An innovative frame design for easier and touch-up free installation. Deco frame comes with pre-drilled holes and can be installed with screws that will be hidden under the inserts.*

Z Frame

- IM only
- Standard deduction = $\frac{1}{16}$ " (all sides for window size)



Note:
For 2-sided Z Frame (IM) other
than café shutter,
standard deduction for

Frame Height = $\frac{1}{16}$ "
Panel Height = $\frac{1}{4}$ "

unless specify otherwise.

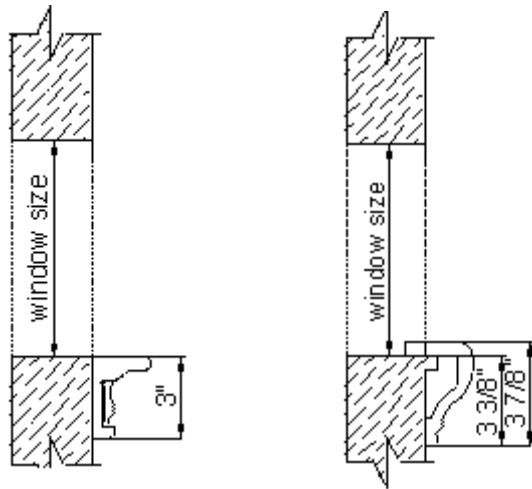
Feature:

- *Z frame is less protruding into the room.*
- *It can help to hide an imperfect window.*
- *The elegant frame profile adds a decorative touch to an ordinary window.*

Measuring For Different Frames

Deco Sill Frame/Designer Sill Cap

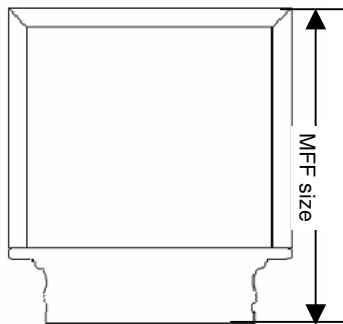
- Window Size Measurement:



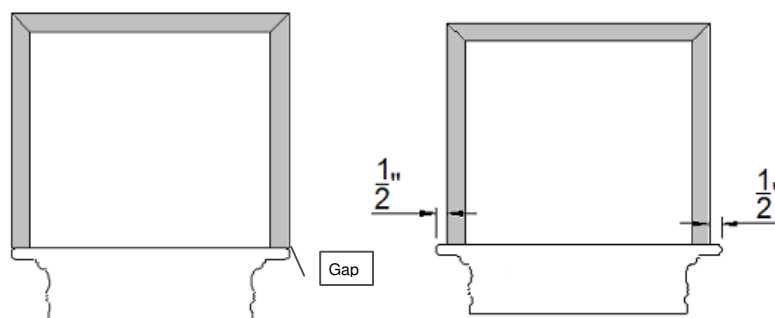
* If extension is requested behind regular frames, then extension will also be added behind sill frame.

- Max Frame to Frame Measurement:

◇ Measure height from the top of top frame to bottom of Deco Sill or Designer Sill Cap.



◇ To eliminate the gap between the side frame and the bullnose edge of the sill frame (see drawings below), sill frame will be extended $\frac{1}{2}$ " on both sides by default or 1" (optional).



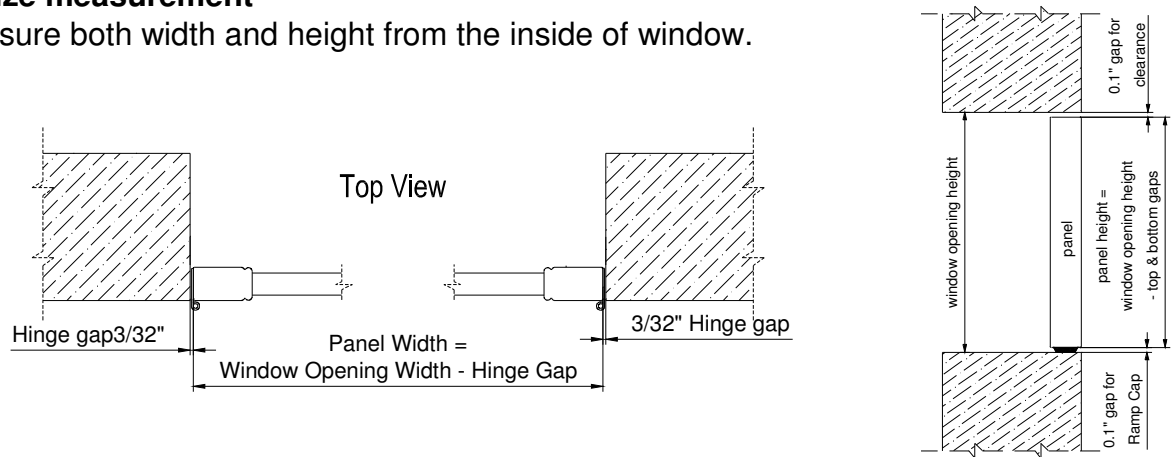
Measuring For Different Frames

Direct Mount

- Shutters are directly hinged to window jambs.
- Hardware, including self mortised hinges and magnets, are mounted onto the panels
- Comes with 4 light blocks, with magnet catches installed on top/bottom light blocks.
- For hinge, part goes on the panel is pre-installed while part goes on the window jamb and hinge pin are included in the hardware box.
- Astragal stile is highly recommended in this mounting option to minimize light gap between panels.
- For shutters used as a door or without bottom support, please refer to the General Product Specification section for details.

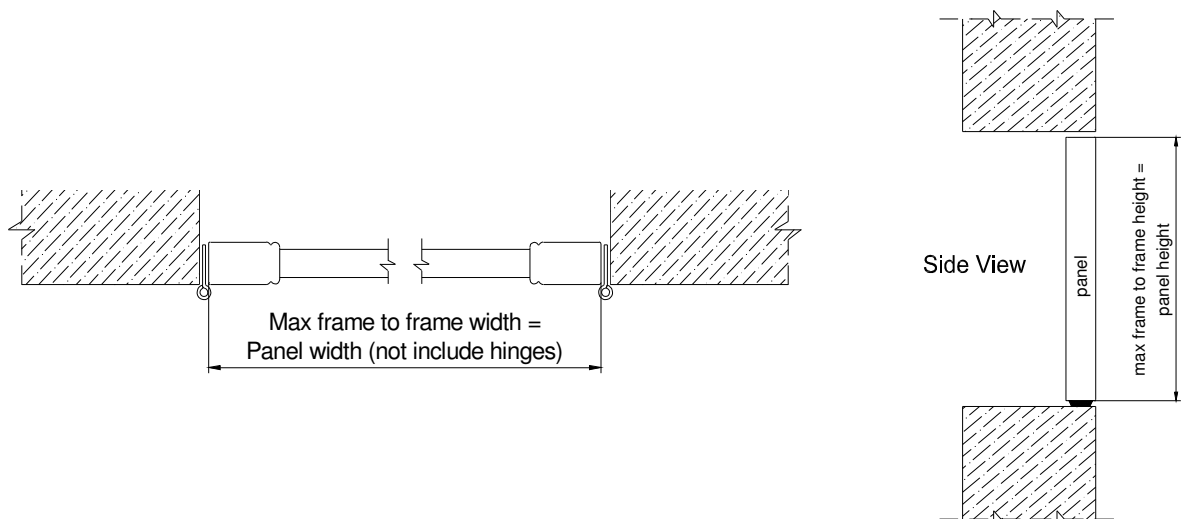
Window Size measurement

- Measure both width and height from the inside of window.



Max Frame to Frame measurement

- Equal to panel size.
- Please note that the Max Frame to Frame width should not include hinge between panel and window jamb.



Measuring For Café Shutters

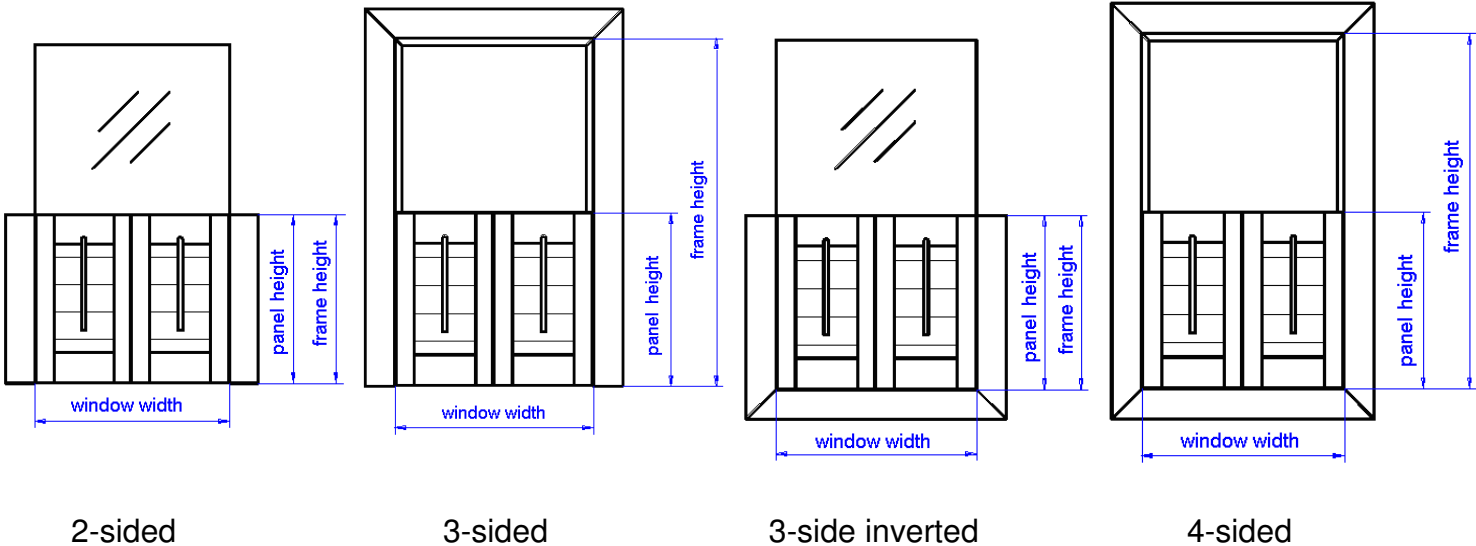
Café Shutters

- The top of the panel requires special finish. Please indicate “café shutter” when placing orders.
- Shutter does not come with recessed magnet on top.

WS = Window Size
MFF = Max Frame to Frame

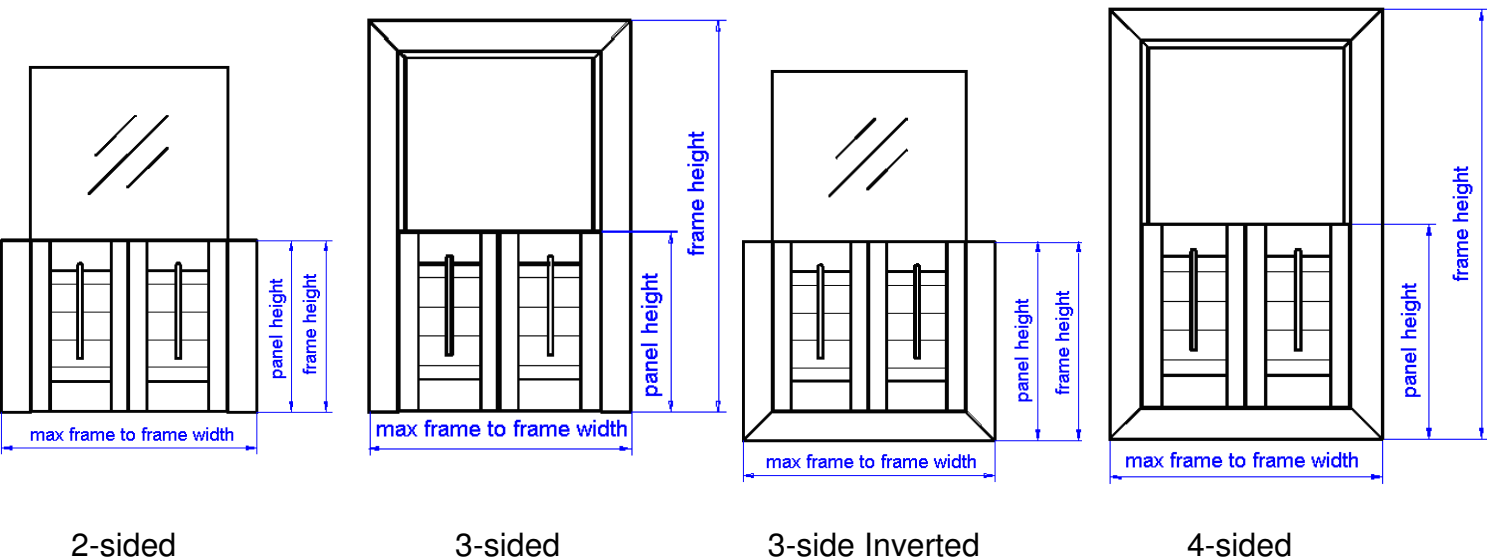
Window size measurement

Panel Height: From bottom of window to top of panels



Max frame to frame size measurement

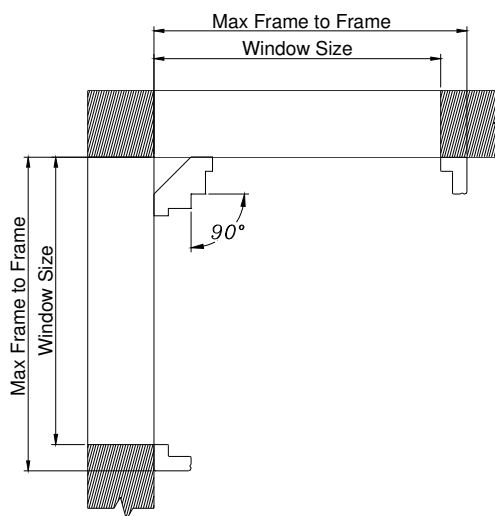
Panel Height: From bottom of frame to top of panels



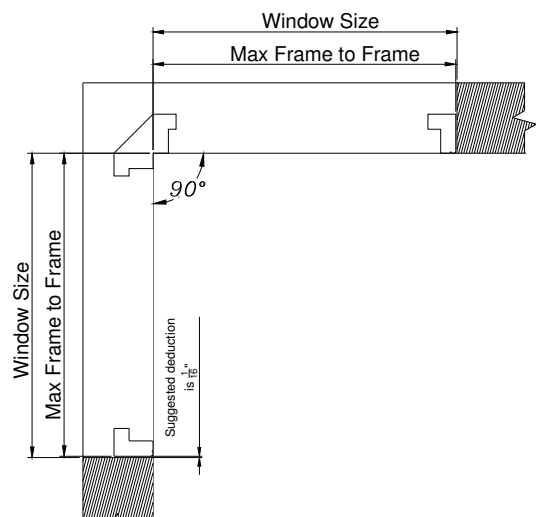
Measuring For Corner Window

Corner Window

- Shutter is made as a single unit with corner post to cover both window
- Standard angle = 90° (For other angles, please refer to Bay Window section)
- Width will be different at different depths in window. To get an accurate measurement, please decide **where frames will be mounted on** and then measure width at correct place as shown in the corner window drawings.



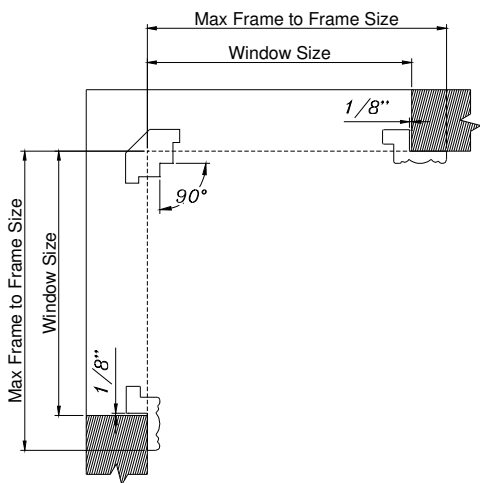
L Frame Outside Mount
(Corner window with a break)



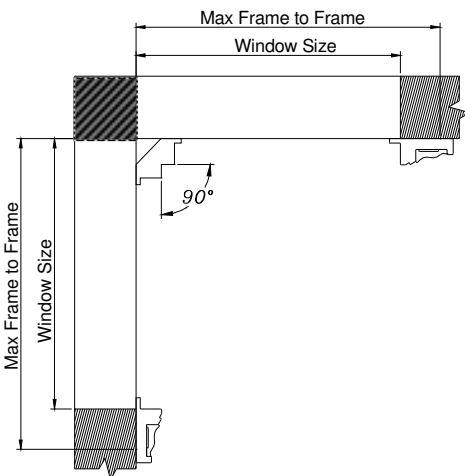
L Frame Inside Mount
(Continuous Corner Window)

Measuring For Corner Window

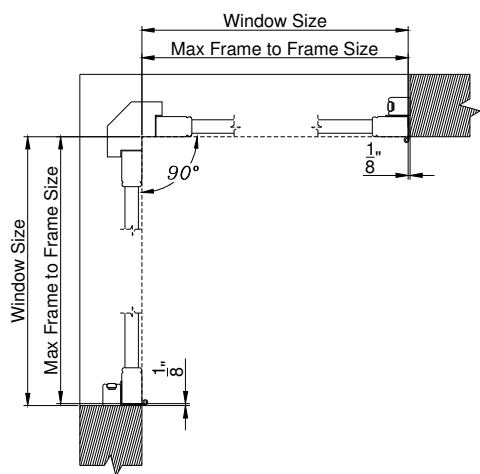
Corner Window (con't)



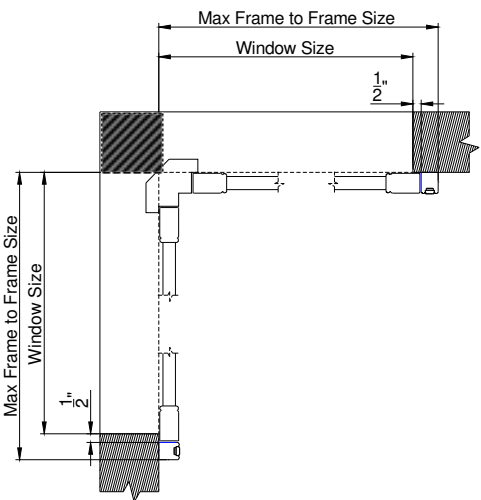
Z Frame Inside Mount
(Continuous Corner Window)



Deco Frame Outside Mount
(Corner Window with a Break)



Hang Strip Inside Mount
(Continuous Corner Window)

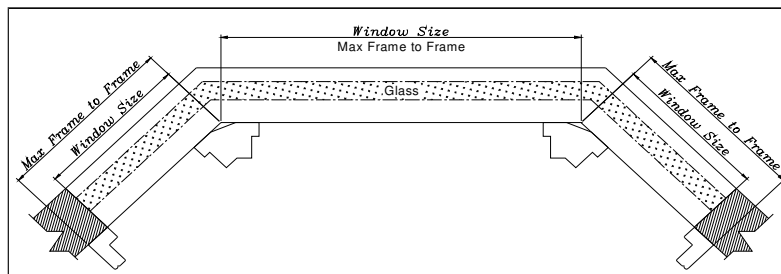


Hang Strip Outside Mount
(Corner Window with a Break)

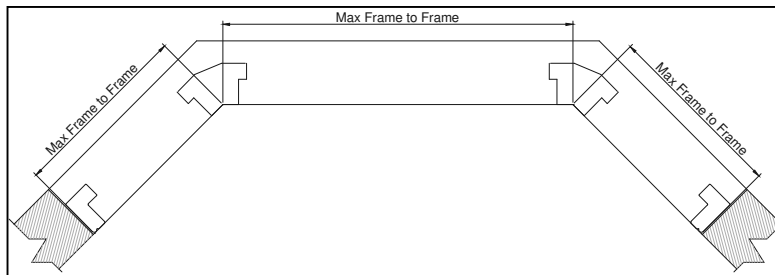
Measuring For Bay Window

Bay Window

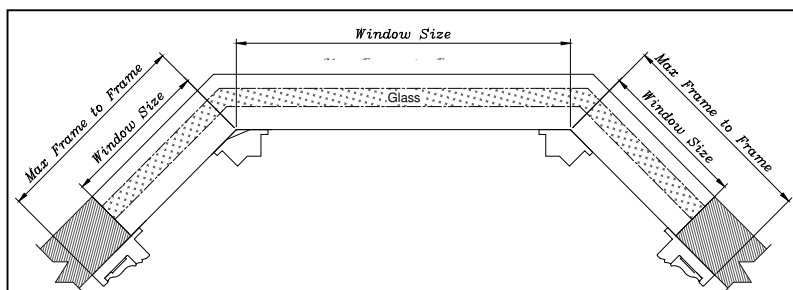
- Shutter is made as a single unit with bay post to cover both window
- Standard angle = 135° (Custom angle is available)
- Width will be different at different depths in window. To get an accurate measurement, please decide where frames will be mounted on and then measure width at correct place as shown following diagrams.



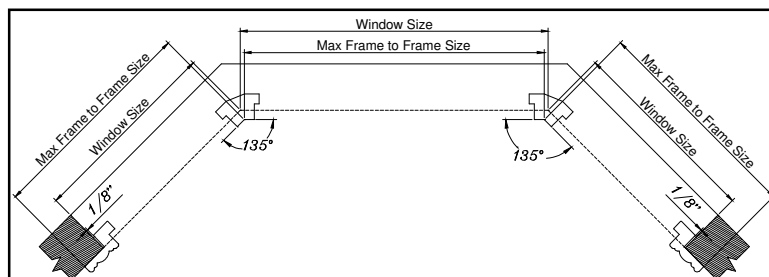
L Frame Outside Mount



L Frame Inside Mount



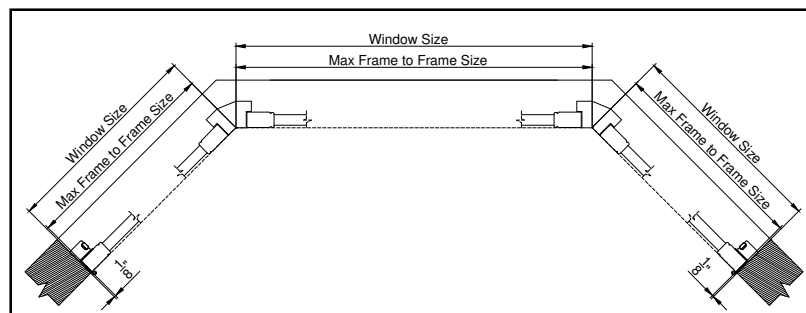
Deco Frame Outside Mount



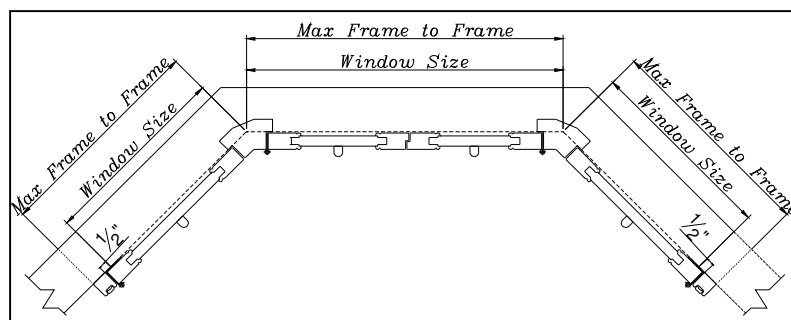
Z Frame Inside Mount

Measuring For Bay Window

Bay Window (con't)



Hang Strip Inside Mount

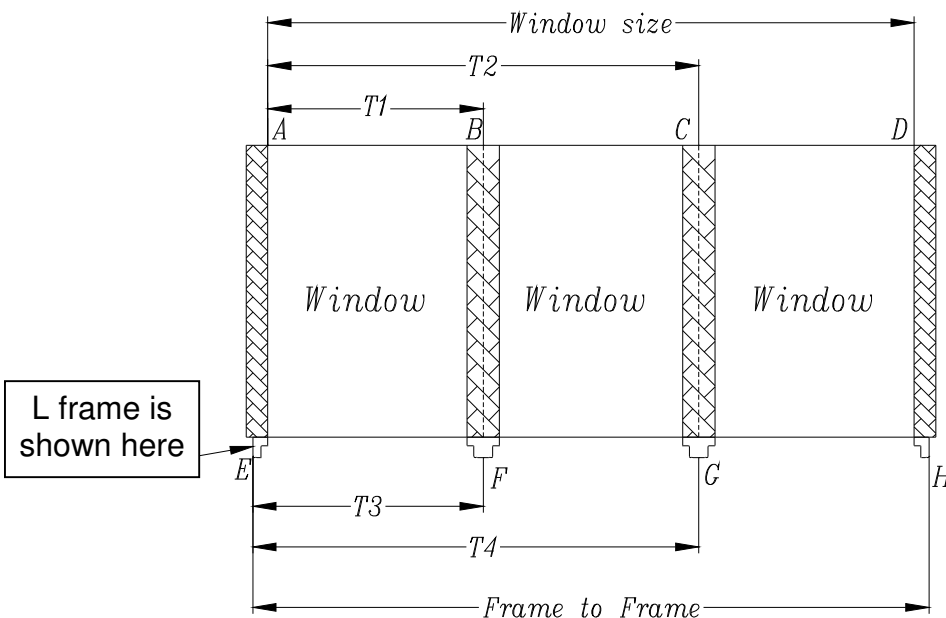


Hang Strip Outside Mount

Measuring For T Post

T Post

- T Post can be used to
 - ◇ divides window vertically to match window design with mullion
 - ◇ accommodate wider window opening/window with mullion
 - allows shutters to be operated individually for better privacy and light control
- Custom T Posts are available to accommodate wider mullions and can be used with any frame type.



Measuring T Post location for Window Size

1. T1 = measure point A to point B
2. T2 = measure point A to point C
3. Measure from point A to point D for the overall window size.

Measuring T Post location for Max Frame to Frame

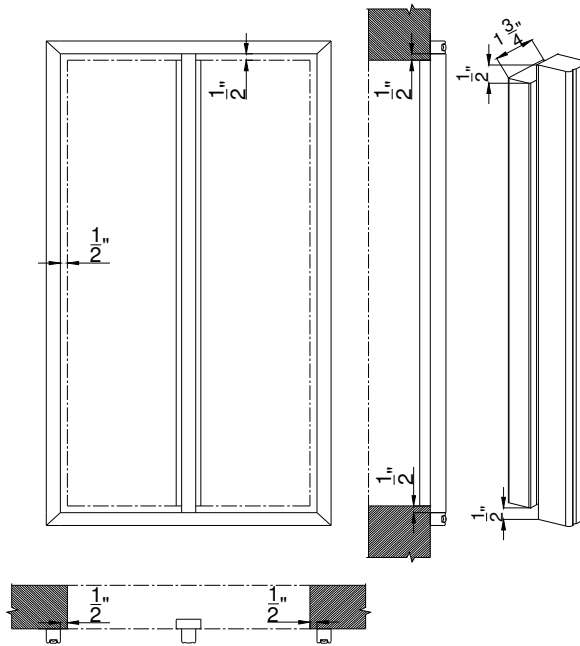
1. T3 = measure point E to point F
2. T4 = measure point E to point G
3. Measure from point E to point H for overall Max Frame to Frame size.

Measuring For T Post

T Post for Hang Strip

OM

- Window size measurement:
 - ◇ T Post is notched in order to fit into the window opening. T Post front face is flushed with hang strip front face.
- Max Frame to Frame measurement:
 - ◇ T Post will not be notched as the distance of the hang strip from the window edge cannot be determined.

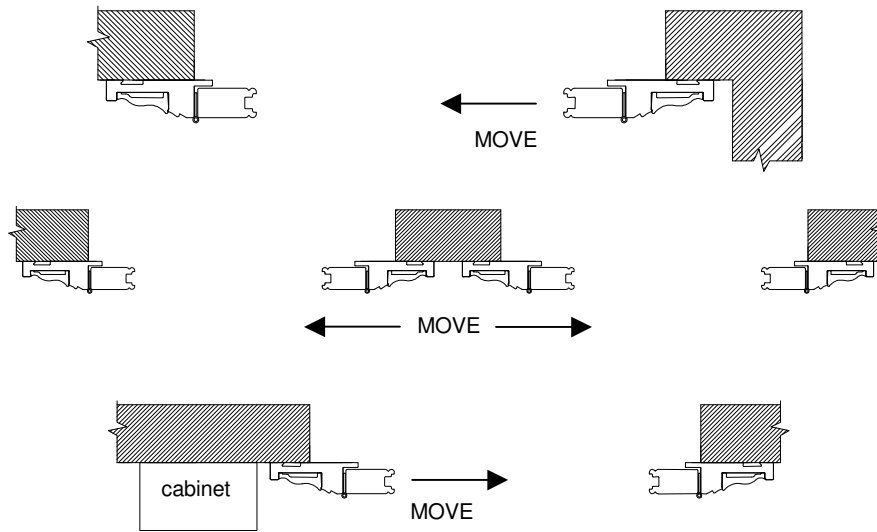


Measuring For Other Window Situation

Other Window Situation

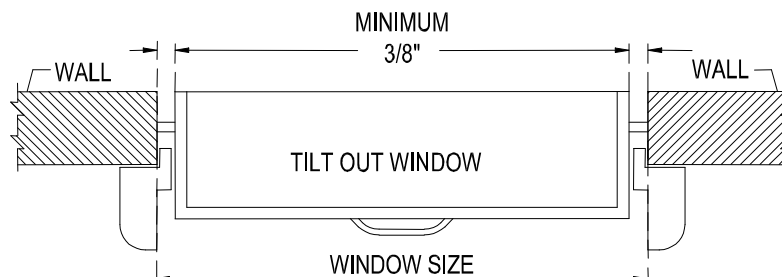
When there is an **obstruction**:

- For the example of a side-by-side window, or when there is obstruction such as a cabinet or wall, Max Frame to Frame measurement is preferred.



For windows with **protruding cranks**:

- Tilt out Z frame and Notch out L frame can be used for windows with protruding cranks or other obstructions.
- Tilt out Z frame requires less depth, leaving more room for tilt out windows to operate.



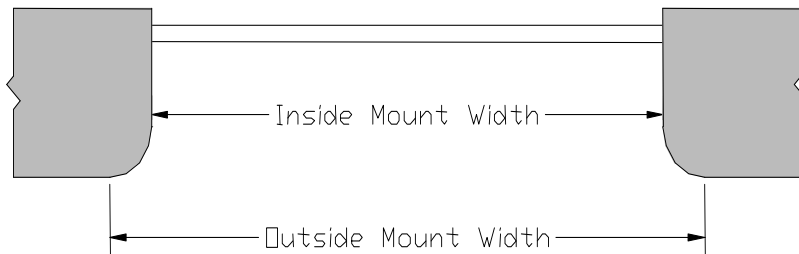
All drawings above are shown in Deco Frame

Measuring For Other Window Situation

Other Window Situation

When there is a **bullnose wall**:

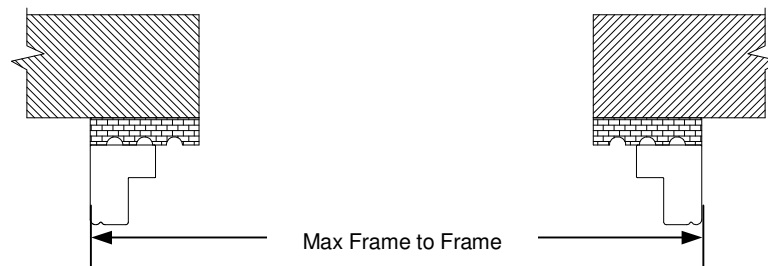
- ◇ Place frame on the flat part of the wall (where the curve ends) for both OM and IM width as shown below.
- ◇ Please provide the smallest measurement for IM.



*** Please measure from the point where bullnose ends and flat wall begins.

For windows with **molding, casing or trim**:

- ◇ Frame can be placed on the molding or beside the molding (see drawing below).
- ◇ Please provide Max Frame to Frame measurement.



Drawings are shown in L frame

Template Requirements

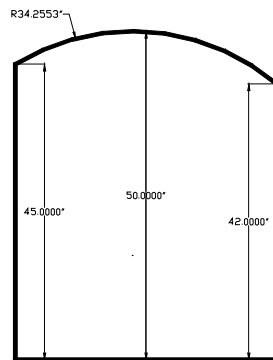
A template is necessary if the specialty shutter is

- Inside mounted
- Will be mounted onto existing molding
- Imperfect arch
- An oval
- Not listed in the program binder

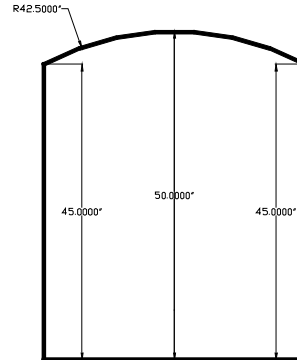
Arches

1) Perfect Arch

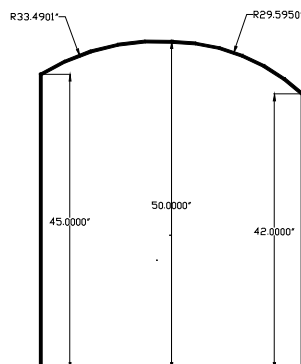
When the same radius with same center point passes through all points on the arch.
Any specialty arch shutter order without a template will be made as a perfect arch.



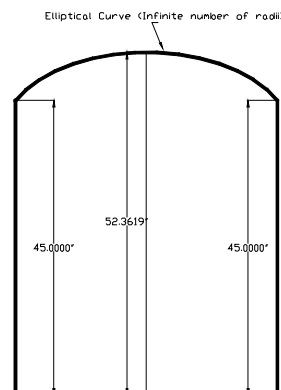
Asymmetrical



Symmetrical



Asymmetrical



Symmetrical

Template Requirements

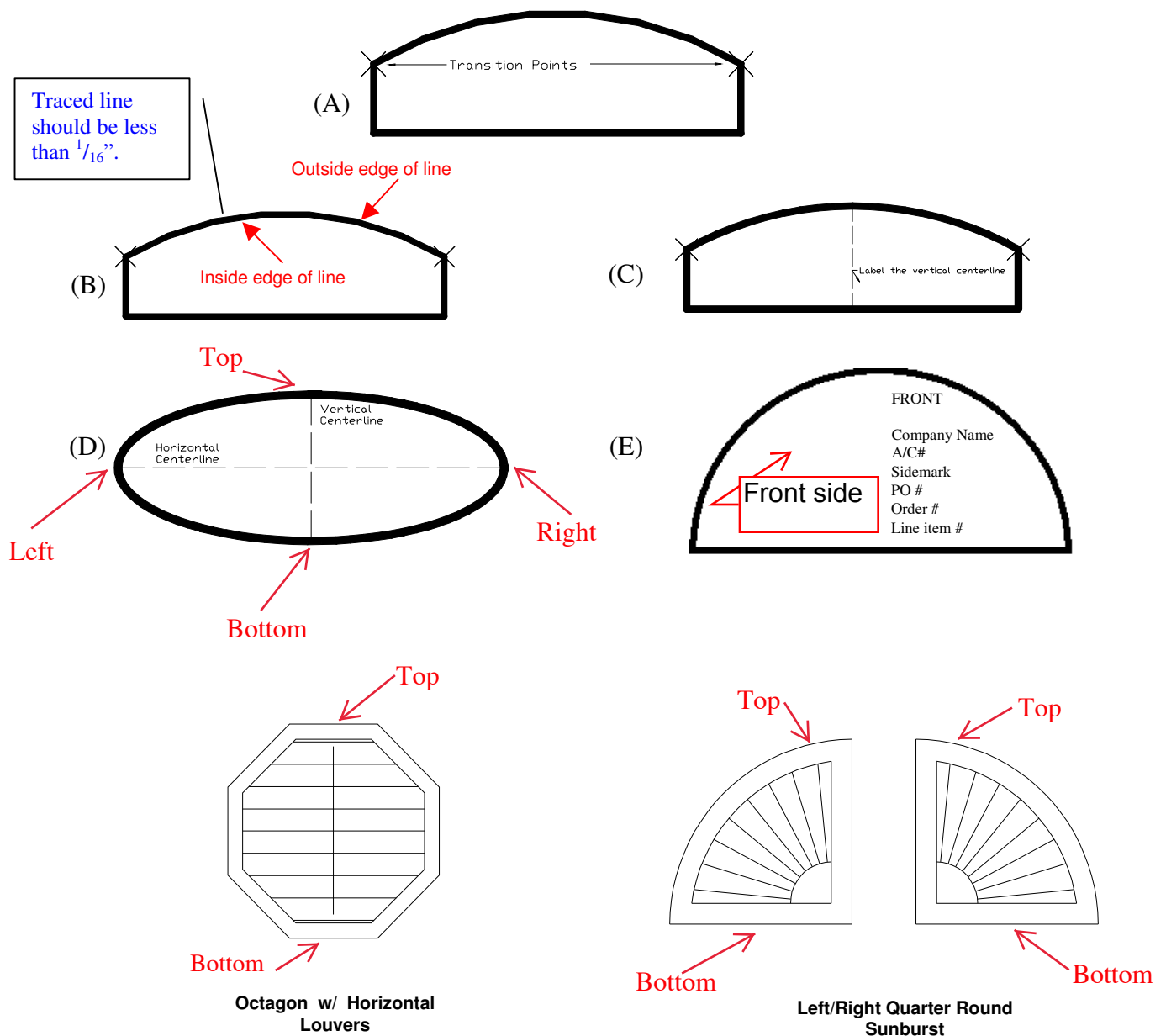
Creating a template

- We recommend using a thick stock paper such as Builders Paper which can be found at your local Home Center/Hardware Store.
- Thinner paper may misrepresent the size and shape of the window due to wrinkles and creases.
- If one sheet of paper does not cover the entire arched area, you may tape multiple sheets together before attaching paper to the window.
- Make sure paper is large enough to extend beyond the arch's width and height
- Paper should cover below the transition points, where the top part of the window ends and the bottom of the arch begins
- Use painter's tape to secure the paper to the wall as paint on the wall may peel if you use regular tape.
- Most arches can be outlined from the front. However, if necessary, you can cut a hole in the center of the paper and outline the arch from the back of the paper through the hole.
- The traced line should not be thicker than 1/16". Thick lines may misrepresent the actual size and shape of the opening. Please mention whether the outside edge or inside edge of line should be used for the shutters. If it is not mentioned then the outside edge of the line will be used for Outside mount orders and the inside edge of line will be used for Inside mount orders.
- Do not draw multiple lines. Make sure horizontal lines are horizontal and not slanted.
- Always mark the transition points as well as draw and label the vertical centerline (divides arch in equal halves) on the paper. For circle or oval shapes, draw and label the horizontal and vertical centerlines.
- For inside mounted L frames, hang strips or direct mounts, please specify on the template whether the template measurements provided are window size or max frame to frame.
- On all templates mark the front side "Front" (facing the inside of the room). All circles, ovals, quarter arches (left or right), octagons and hexagons also need to have the sides marked "Top" and "Bottom" on the corresponding sides of the template.
- **On the template, indicate your Company Name, Customer Account ID#, Side Mark, PO#, Order#, Line item # and "Front side", so that the template can be properly identified.**
- **Please note: the measurements indicated on your template should be the same as those entered into the ordering system. If not, you will have to verify via an on-hold question.**

These requirements should be followed closely as incomplete templates may prolong lead time

Template Requirements

Marking Examples



Template Requirements

Sending Template

- By using couriers services such as FedEx and DHL, templates are usually delivered to China within 5 to 7 business days.
- Choose a shipping method that has tracking ability. Packages with content value greater than \$10.00 are subjected to customs duty.
- **It is not recommended to send templates via the United States Postal Service (USPS) as this will lead to extremely long delays.**
- If the customs duty is not cleared or paid by the customer, the delivery process will be delayed. It is advised to write: **PAPER DOCUMENT ENCLOSED, NOT FOR COMMERCIAL RESALE** to avoid unnecessary custom duty.
- A CAD drawing will be sent to you once we receive your template. The lead-time begins after we receive your final drawing confirmation.

Send your template to:

Attn: Liu Ya Yi (Shutter Factory)
Company Name: NienMade (Dong Guan) Window Fashions Co., Ltd
Address: No. 1, Zhao Feng Industrial South Road,
QiShi Town, DongGuan City, GuangDong Province
China 523517
Tel: +86 769 86716666



NORMAN®
SHUTTERS. BLINDS. SHADES.

SHUTTERS LIFETIME LIMITED WARRANTY



Norman® shutters are backed by a Lifetime Limited Warranty against original defects in materials or workmanship for as long as the original purchaser owns the shutter(s), provided that such products were properly installed. Any resale or other transfer of the product and/or materials voids this warranty.

Composite shutters have a Lifetime Limited warranty covering colorfastness and warping. Painted and stained shutters have a 3 year warranty covering colorfastness. Wood materials may warp in high humidity areas. The manufacturer considers this normal wear and tear and is not covered by this warranty.

Motorization components are covered for two years from the date of purchase.

To Obtain Warranty Service

- Please contact the original dealer for warranty assistance
- For product manuals and technical support visit NormanProductWarranty.com/installation-quick.asp

Repair or remake of defective products is the sole remedy under this warranty and in no event shall Norman® or its licensed distributors or dealers be liable for costs to remove and/or reinstall the product. Repairs will be made with like or similar parts. This warranty does not include shipping charges, costs of labor for measuring, installation, or trip charges. Warranty claims must be accompanied by proof of purchase, as well as details regarding the nature of the problem, location of the product, etc.

Please Note:

This warranty does not cover any conditions caused by normal wear and tear, unauthorized repairs, accidents, alterations, misuse, abuse, exposure to the elements, excessive humidity, fading, failure to follow instructions for measurement, installation, cleaning or maintenance.

This warranty is exclusive and in lieu of all other obligations, liabilities or warranties. All other warranties, both expressed and implied, are explicitly disclaimed. In no event shall Norman Window Fashions® or Norman® distributors/dealers be liable or responsible for incidental or consequential damages, or for any other direct or indirect damage, loss, cost, expense or fee. Some states do not allow the exclusion or limitation of implied warranties, or incidental or consequential damages, so the above limitation may not apply to your state.

Exceptions:

Different warranty periods and terms apply for commercial products and applications.